

# Diagnosing Index-Shift and Convergence-Condition Errors

## Answer Key

### Precalculus

#### Quick Answers

---

- |                                  |                                                               |
|----------------------------------|---------------------------------------------------------------|
| 1. (a) the correct sum = 20, —   | 6. (a) the correct sum = $\frac{20}{3}$ , —                   |
| 2. (a) the correct sum = 120, —  | 7. (a) the term at $n = 10 = \frac{10}{11}$ , (a) the term at |
| 3. (a) the convergent sum = 3, — | $n = 11 = -\frac{11}{12}$ , —                                 |
| 4. (a) the correct limit = 3, —  | 8. (a) the correct sum = $\frac{3}{2}$ , —                    |
| 5. (a) the correct sum = 55, —   |                                                               |
- 

#### What is verified

---

9 of 17 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

---

#### Curriculum

---

**Level:** Precalculus

HSF-BF.A.2 – *problems 1, 2, 3, 5, 6, 8*

LIM-1 – *problems 4, 7*

*Difficulty 1–5 of 5 across 17 tagged checks.*

---

#### Common wrong answers

---

- If they got 21: used the sum-from-1 formula and never removed the  $k = 1$  term*
  - If they got 363: moved the lower limit down by one but left the upper limit alone, adding a fifth term*
  - If they got  $-2$ : applied  $a/(1 - r)$  to a ratio larger than 1, producing a negative sum for a series of positive terms*
  - If they got 1: cancelled the  $n$  terms and read the limit as 1*
  - If they got 140: substituted  $j = k - 2$  in the term but kept the old upper limit of 7*
  - If they got 1.67: read the first term as  $5/4$  by substituting  $n = 1$  into  $5r^n$  instead of  $5r^{(n-1)}$*
  - If they got 6: used the  $n = 0$  term as  $a$ , ignoring that the series starts at  $n = 2$*
- 

1. A student evaluates  $\sum_{k=2}^6 k$  with  $\frac{n(n+1)}{2}$  at  $n = 6$  and writes 21. (a) Correct value. (b) Name the error.

**Solution (a):**  $2 + 3 + 4 + 5 + 6 = 20$ , or  $\frac{6 \cdot 7}{2} - 1 = 21 - 1$ .

|    |
|----|
| 20 |
|----|

**Solution (b) — instructor-judged.** The formula  $\frac{n(n+1)}{2}$  sums from  $k = 1$ , but this sum starts at  $k = 2$ . The  $k = 1$  term must be subtracted, so the student's answer is exactly 1 too large. Full credit requires naming the starting index as the problem.

---

2. A student rewrites  $\sum_{n=1}^4 3^n$  as  $\sum_{m=0}^4 3^{m+1}$  and gets 363. (a) Correct value. (b) Name the error.

**Solution (a):**  $3 + 9 + 27 + 81 = 120$ .

120

**Solution (b) — instructor-judged.** The substitution  $m = n - 1$  lowers *both* limits by 1: the sum should be  $\sum_{m=0}^3 3^{m+1}$ . Leaving the upper limit at 4 adds a fifth term,  $3^5 = 243$ , and indeed  $120 + 243 = 363$ . Full credit requires the statement that both limits shift by the same amount.

---

3. A student applies  $\frac{a}{1-r}$  to  $\sum_{n=0}^{\infty} \left(\frac{3}{2}\right)^n$  and reports  $-2$ . (a) Evaluate  $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$ . (b) Why does the formula fail on the student's series?

**Solution (a):** Here  $r = \frac{2}{3}$  and  $|r| < 1$ , and the first term is 1, so the sum is  $\frac{1}{1 - 2/3} = \frac{1}{1/3}$ .

3

**Solution (b) — instructor-judged.** The formula  $\frac{a}{1-r}$  is valid only when  $|r| < 1$ . With  $r = \frac{3}{2}$  the terms  $1, 1.5, 2.25, \dots$  grow without bound, so the series diverges and has no sum. The negative answer is the giveaway: a series of positive terms can never sum to a negative number, so the formula was used outside its domain. Full credit requires the convergence condition stated.

---

4. A student says  $\lim_{n \rightarrow \infty} \frac{3n+2}{n-5} = 1$ . (a) Correct limit. (b) Name the error.

**Solution (a):** Divide numerator and denominator by  $n$ :

$$\frac{3n+2}{n-5} = \frac{3 + 2/n}{1 - 5/n} \rightarrow \frac{3+0}{1-0}$$

3

**Solution (b) — instructor-judged.** The constants 2 and  $-5$  do become negligible, but the *coefficients* 3 and 1 do not cancel — the numerator grows three times as fast as the denominator. The limit is the ratio of the leading coefficients, 3, not 1. Full credit requires distinguishing the constants from the coefficients.

---

5. Rewriting  $\sum_{k=3}^7 (k-2)^2$  with  $j = k-2$ , a student writes  $\sum_{j=1}^7 j^2 = 140$ . (a) Correct value. (b) Name the error and the term count.

**Solution (a):**  $1 + 4 + 9 + 16 + 25 = 55$ .

55

**Solution (b) — instructor-judged.** With  $j = k - 2$ ,  $k = 3$  gives  $j = 1$  and  $k = 7$  gives  $j = 5$ , so the correct rewrite is  $\sum_{j=1}^5 j^2$ . The sum has  $7 - 3 + 1 = 5$  terms before the shift and must still have 5 after it. The student kept the old upper limit, adding 36 and 49:  $55 + 36 + 49 = 140$ . Full credit requires the corrected upper limit and the term count.

**6.** A student evaluates  $\sum_{n=1}^{\infty} 5 \left(\frac{1}{4}\right)^{n-1}$  as  $\frac{5/4}{1-1/4} \approx 1.67$ . (a) Correct sum. (b) Name the error in  $a$ .

**Solution (a):** The first term is the  $n = 1$  term,  $5 \left(\frac{1}{4}\right)^0 = 5$ , and  $r = \frac{1}{4}$ , so the sum is  $\frac{5}{1 - 1/4} = \frac{5}{3/4}$ .  $\frac{20}{3}$

**Solution (b) — instructor-judged.** The exponent is  $n - 1$ , so substituting  $n = 1$  gives  $\left(\frac{1}{4}\right)^0 = 1$  and the first term is 5, not  $\frac{5}{4}$ . The student read the term as if the exponent were  $n$ . The value of  $a$  is always the series' own first term as written. Full credit requires the exponent  $n - 1$  named as the cause.

**7.** A student says  $a_n = (-1)^n \frac{n}{n+1}$  converges to 1. (a) Compute  $a_{10}$  and  $a_{11}$ . (b) Name the error.

**Solution (a):**  $a_{10} = (+1)\frac{10}{11}$ , so

$$a_{10} = \frac{10}{11}$$

and  $a_{11} = (-1)\frac{11}{12}$ , so

$$a_{11} = -\frac{11}{12}$$

**Solution (b) — instructor-judged.** The size  $\frac{n}{n+1}$  does approach 1, but the factor  $(-1)^n$  flips the sign at every step, so consecutive terms sit near  $+1$  and near  $-1$  forever. The sequence never settles on one number: it diverges, and no limit exists. Full credit requires the alternating sign as the reason and the conclusion that there is no limit.

**8.** A student evaluates  $\sum_{n=2}^{\infty} 3 \left(\frac{1}{2}\right)^n$  as  $\frac{3}{1-1/2} = 6$ . (a) Correct sum. (b) Name the error and give the right  $a$ .

**Solution (a):** The first term of this series is the  $n = 2$  term,  $3 \left(\frac{1}{2}\right)^2 = \frac{3}{4}$ , with  $r = \frac{1}{2}$ :

$$\frac{3/4}{1 - 1/2} = \frac{3/4}{1/2}$$

$$\frac{3}{2}$$

**Solution (b) — instructor-judged.** In  $\frac{a}{1-r}$ ,  $a$  is the first term of the series *as written*. This series starts at  $n = 2$ , so  $a = \frac{3}{4}$ ; the student used the  $n = 0$  term, 3, which belongs to a longer series. Adding the missing  $n = 0$  and  $n = 1$  terms,  $3 + \frac{3}{2} = \frac{9}{2}$ , to the correct  $\frac{3}{2}$  recovers the student's 6 — which is exactly the sum of the wrong series. Full credit requires  $a = \frac{3}{4}$  and the reason.